



5/4/2025

Hospital - Case Study

DataBase Assignment 3

Team members:

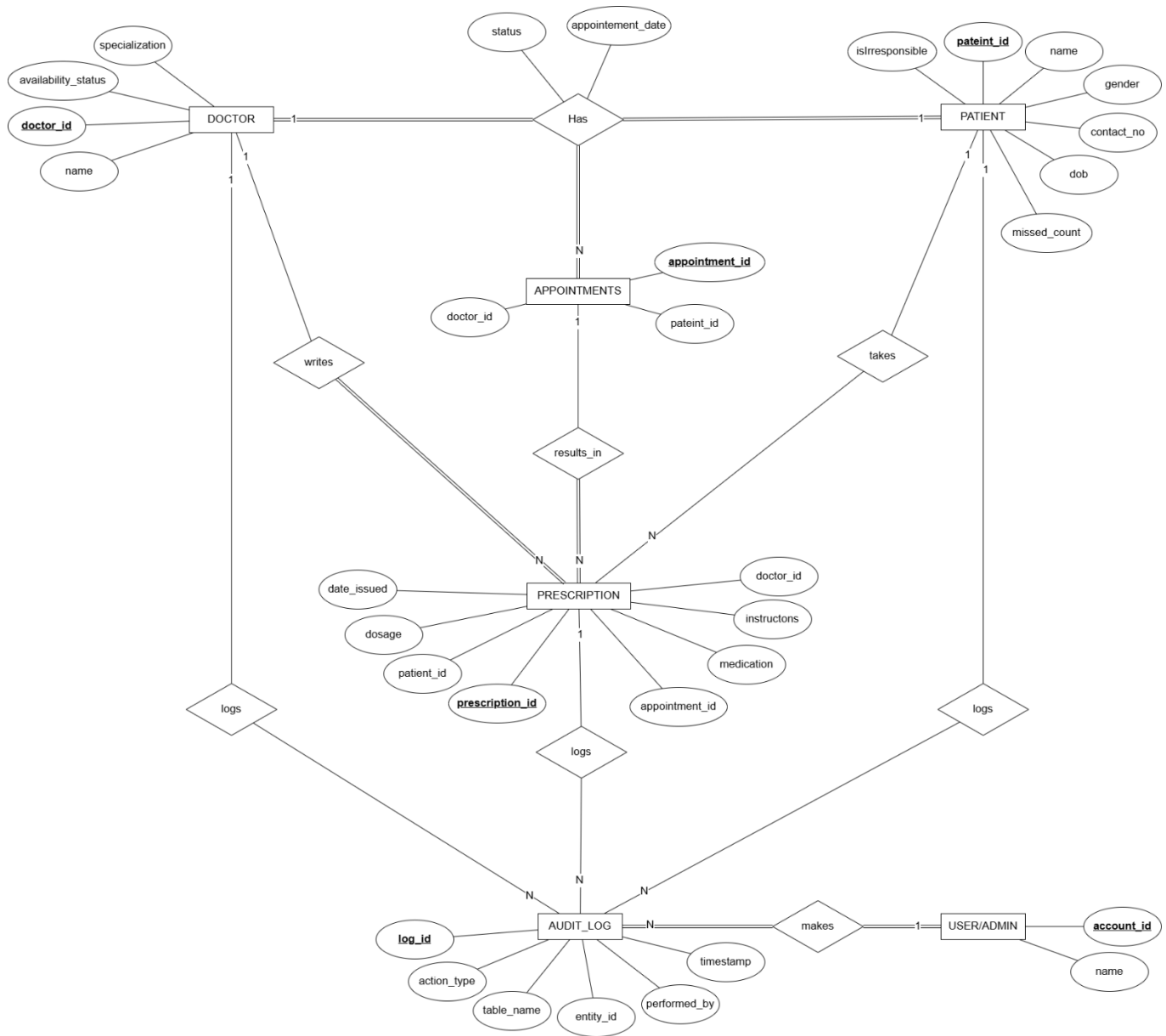
Moiz Ansari [23i-0523]

Abdullah Siddiqui [23i-0617]

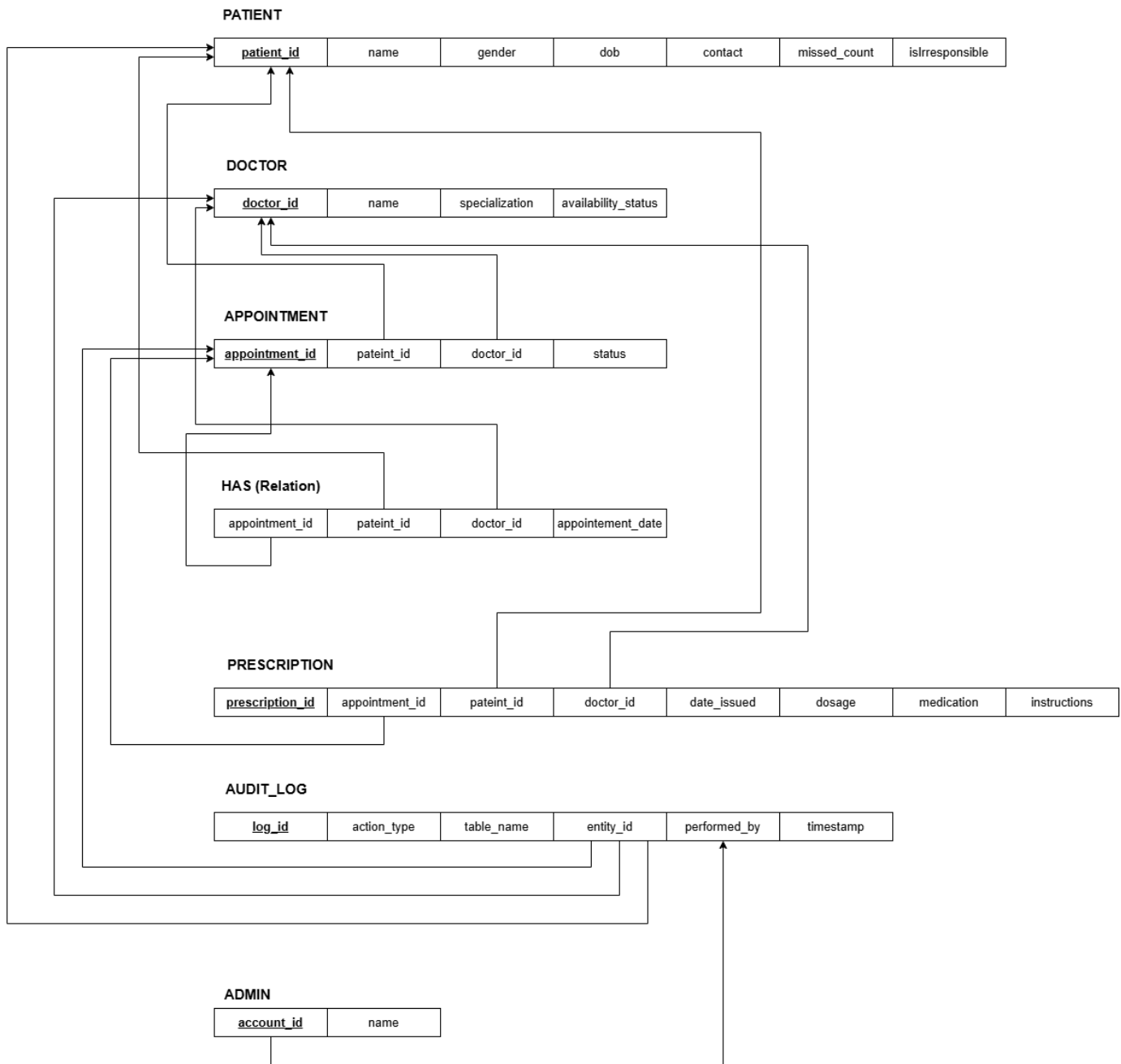
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ERD



Relational Schema



Modifications

- Added columns missed_count and isIrresponsible (flag) in PATIENTS table for missing appointments.
- Used Appointment table as Has(relation) table.

- Made table of Admin and added account_no and names.

Trigger Testing

TASK 1

Upon insertion of patient values, Audit Logs following data.

```
--Task 1
Create Trigger patient_safety
on PATIENT
After Insert
as
begin
    Declare @temp INT;
    Select @temp=patient_id
    from inserted
    DECLARE @pres_id INT
    SET @pres_id = (SELECT TOP 1 pr.log_id FROM AUDIT_LOG pr ORDER BY pr.log_id DESC)
    SET @pres_id = @pres_id + 1;
    Insert into audit_log values(@pres_id,'INSERT','PATIENT',@temp, 'admin1',getdate());
end

INSERT INTO PATIENT VALUES(1000,'Ali','Male','2010-05-03','03350215325');
SELECT * FROM AUDIT_LOG
```

100 %

Results Messages

	log_id	action_type	table_name	entity_id	performed_by	timestamp
1	1	INSERT	PATIENTS	101	admin1	2025-04-24 14:38:00.000
2	2	BOOK	APPOINTMENTS	301	admin1	2025-04-24 14:38:00.000
3	3	BOOK	APPOINTMENTS	302	admin1	2025-04-24 14:38:00.000
4	4	CANCEL	APPOINTMENTS	303	admin2	2025-04-24 14:38:00.000
5	5	INSERT	PATIENT	1000	admin1	2025-05-04 16:24:41.790

TASK 2

Upon insertion and deletion, audit logs respective message like “BOOK” on insertion and “CANCEL” on deletion.

```
INSERT INTO APPOINTMENTS VALUES(3000,1000,201,'2025-02-03','ACTIVE');  
DELETE APPOINTMENTS  
WHERE appointment_id=3000  
SELECT * FROM AUDIT_LOG
```

100 %

Results Messages

	log_id	action_type	table_name	entity_id	performed_by	timestamp
1	1	INSERT	PATIENTS	101	admin1	2025-04-24 14:38:00.000
2	2	BOOK	APPOINTMENTS	301	admin1	2025-04-24 14:38:00.000
3	3	BOOK	APPOINTMENTS	302	admin1	2025-04-24 14:38:00.000
4	4	CANCEL	APPOINTMENTS	303	admin2	2025-04-24 14:38:00.000
5	5	INSERT	PATIENT	1000	admin1	2025-05-04 16:24:41.790
6	6	BOOK	APPOINTMENT	3000	admin1	2025-05-04 16:29:59.253
7	7	CANCEL	APPOINTMENT	3000	admin1	2025-05-04 16:30:51.863

TASK 3

When changing availability from Unavailable to Available, it logs status update data. It would not log if the same status is changed to previous value.

```
UPDATE DOCTOR  
SET availability_status='Available'  
WHERE doctor_id=202  
SELECT * FROM AUDIT_LOG
```

100 %

Results Messages

	log_id	action_type	table_name	entity_id	performed_by	timestamp
1	1	INSERT	PATIENTS	101	admin1	2025-04-24 14:38:00.000
2	2	BOOK	APPOINTMENTS	301	admin1	2025-04-24 14:38:00.000
3	3	BOOK	APPOINTMENTS	302	admin1	2025-04-24 14:38:00.000
4	4	CANCEL	APPOINTMENTS	303	admin2	2025-04-24 14:38:00.000
5	5	INSERT	PATIENT	1000	admin1	2025-05-04 16:24:41.790
6	6	BOOK	APPOINTMENT	3000	admin1	2025-05-04 16:29:59.253
7	7	CANCEL	APPOINTMENT	3000	admin1	2025-05-04 16:30:51.863
8	8	STATUS_UPDATE	DOCTOR	202	admin1	2025-05-04 16:34:18.813

TASK 4

Upon adding an appointment for an unavailable doctor, it stored the rejected insert record. It will do the same if same appointment is done again with already existing appointment date for a patient.

```
INSERT INTO APPOINTMENTS VALUES(5000,1000,202,'2025-03-04','PENDING');
SELECT * FROM AUDIT_LOG
```

	log_id	action_type	table_name	entity_id	performed_by	timestamp
1	1	INSERT	PATIENTS	101	admin1	2025-04-24 14:38:00.000
2	2	BOOK	APPOINTMENTS	301	admin1	2025-04-24 14:38:00.000
3	3	BOOK	APPOINTMENTS	302	admin1	2025-04-24 14:38:00.000
4	4	CANCEL	APPOINTMENTS	303	admin2	2025-04-24 14:38:00.000
5	5	INSERT	PATIENT	1000	admin1	2025-05-04 16:24:41.790
6	6	BOOK	APPOINTMENT	3000	admin1	2025-05-04 16:29:59.253
7	7	CANCEL	APPOINTMENT	3000	admin1	2025-05-04 16:30:51.863
8	8	STATUS_UPDATE	DOCTOR	202	admin1	2025-05-04 16:34:18.813
9	9	STATUS_UPDATE	DOCTOR	202	admin1	2025-05-04 16:40:33.637
10	10	REJECTED_INSERT	APPOINTMENT	5000	admin1	2025-05-04 16:41:14.973

TASK 5

Added prescription

```
--
UPDATE APPOINTMENTS SET status = 'Completed' WHERE appointment_id = 301;
SELECT * FROM AUDIT_LOG
```

	log_id	action_type	table_name	entity_id	performed_by	timestamp
7	7	PRESCRIBE	PRESCRIPTIONS	404	23	2025-05-04 10:00:00.000
8	8	MISSED	APPOINTMENTS	309	23	2025-05-04 10:00:00.000
9	9	INSERT	PATIENTS	121	17	2025-05-04 17:54:14.200
10	10	BOOK	APPOINTMENTS	321	17	2025-05-04 17:57:30.727
11	11	BOOK	APPOINTMENTS	321	23	2025-05-04 17:57:30.733
12	12	CANCEL	APPOINTMENTS	NULL	17	2025-05-04 17:58:22.017
13	13	STATUS_UPDATE	DOCTORS	201	17	2025-05-04 17:58:51.807
14	14	REJECTED_INSERT	APPOINTMENT	321	17	2025-05-04 17:59:19.613
15	15	PRESCRIBE	PRESCRIPTIONS	406	23	2025-05-04 18:00:01.643

Rejected Prescription

--Test 5

DECLARE @app_id INT = 299

SET @app_id = (SELECT TOP 1 ap.appointment_id FROM APPOINTMENTS ap ORDER BY ap.appointment_id DESC)

SET @app_id = @app_id + 1;

INSERT INTO APPOINTMENTS(appointment_id, patient_id, doctor_id, appointment_date, status)

VALUES (@app_id, 102, 201, GETDATE(), 'Booked')

100 %

ResultsMessages

	log_id	action_type	table_name	entity_id	performed_by	timestamp
21	21	MISSED	APPOINTMENTS	305	23	2025-05-04 18:21:38.393
22	22	MISSED	APPOINTMENTS	306	23	2025-05-04 18:21:38.393
23	23	REJECTED_INSERT	APPOINTMENT	323	17	2025-05-04 18:22:10.310
24	24	REJECTED_INSERT	APPOINTMENT	323	17	2025-05-04 18:23:20.783
25	25	REJECTED_INSERT	APPOINTMENT	325	17	2025-05-04 18:41:36.663

TASK 6

INSERT INTO APPOINTMENTS VALUES (322, 101, 201, '2025-05-01', 'Booked');

INSERT INTO APPOINTMENTS VALUES (324, 120, 210, '2025-05-14', 'Booked');

SELECT * FROM AUDIT_LOG

100 %

ResultsMessages

	log_id	action_type	table_name	entity_id	performed_by	timestamp
10	10	BOOK	APPOINTMENTS	321	17	2025-05-04 17:57:30.727
11	11	BOOK			23	2025-05-04 17:57:30.733
12	12	CANCEL	APPOINTMENTS	NULL	17	2025-05-04 17:58:22.017
13	13	STATUS_UPDATE	DOCTORS	201	17	2025-05-04 17:58:51.807
14	14	REJECTED_INSERT	APPOINTMENT	321	17	2025-05-04 17:59:19.613
15	15	PRESCRIBE	PRESCRIPTIONS	406	23	2025-05-04 18:00:01.643
16	16	STATUS_UPDATE	DOCTORS	201	23	2025-05-04 18:00:01.680
17	17	REJECTED_INSERT	APPOINTMENT	322	17	2025-05-04 18:00:58.620
18	18	BOOK	APPOINTMENTS	324	17	2025-05-04 18:00:58.623
19	19	BOOK	APPOINTMENTS	324	23	2025-05-04 18:00:58.627

TASK 7

-- Test 7

```
UPDATE APPOINTMENTS SET status = 'Missed' WHERE appointment_id = 302;  
UPDATE APPOINTMENTS SET status = 'Missed' WHERE appointment_id = 305;  
UPDATE APPOINTMENTS SET status = 'Missed' WHERE appointment_id = 306;  
SELECT * FROM AUDIT_LOG
```

100 %

Results Messages

	log_id	action_type	table_name	entity_id	performed_by	timestamp
16	16	STATUS_...	DOCTORS	201	23	2025-05-04 18:00:01.680
17	17	REJECTE...	APPOINTMENT	322	17	2025-05-04 18:00:58.620
18	18	BOOK	APPOINTMENTS	324	17	2025-05-04 18:00:58.623
19	19	BOOK	APPOINTMENTS	324	23	2025-05-04 18:00:58.627
20	20	MISSED	APPOINTMENTS	302	23	2025-05-04 18:21:38.380
21	21	MISSED	APPOINTMENTS	305	23	2025-05-04 18:21:38.393
22	22	MISSED	APPOINTMENTS	306	23	2025-05-04 18:21:38.393

```
INSERT INTO APPOINTMENTS VALUES (323, 120, 201, '2025-05-01', 'Booked');  
SELECT * FROM AUDIT_LOG
```

100 %

Results Messages

	log_id	action_type	table_name	entity_id	performed_by	timestamp
20	20	MISSED	APPOINTMENTS	302	23	2025-05-04 18:21:38.380
21	21	MISSED	APPOINTMENTS	305	23	2025-05-04 18:21:38.393
22	22	MISSED	APPOINTMENTS	306	23	2025-05-04 18:21:38.393
23	23	REJECTED_INSERT	APPOINTMENT	323	17	2025-05-04 18:22:10.310
24	24	REJECTED_INSERT	APPOINTMENT	323	17	2025-05-04 18:23:20.783

Tables Updated

Results Messages							
	patient_id	name	gender	dob	contact	missed_count	isIrresponsible
1	101	Alice Smith	F	1990-01-01	1234567890	3	1
2	102	Bob Johnson	M	1985-06-15	0987654321	3	1
3	103	Charlie Davis	M	1992-09-20	1122334455	1	0
4	104	David Wilson	M	1988-03-10	2233445566	4	1
5	105	Emma Brown	F	1995-07-22	3344556677	1	0

	doctor_id	name	specialization	availability_status
2	202	Dr. John Adams	Neurology	Unavailable
3	203	Dr. Sarah Lee	Orthopedics	Available
4	204	Dr. Michael Br...	Pediatrics	Available
5	205	Dr. Laura Wils...	Oncology	Unavailable
6	206	Dr. David Kim	Endocrinolo...	Available
7	207	Dr. Anna Patel	Psychiatry	Available